

Oxygen consumption of rainbow trout (*Salmo gairdneri*) in relation to activity and salinity¹

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The oxygen consumption of the rainbow trout ranging in size from 23 to 196 g was measured at 5 °C and 15 °C in fresh water and salinities of 7.5‰, 15‰, and 30‰. The standard rate of metabolism and metabolic rates at various levels of steady swimming were determined.

The cost of swimming and the cost of osmoregulation in terms of oxygen consumption for a 100-g fish were calculated on the assumption that there was no cost of osmoregulation in a salinity of 7.5‰. In fresh water and in 15‰ salinity the cost of osmoregulation was 20% of total metabolism while in a salinity of 30‰ it was 27%. This 27% of the total metabolism attributed to cost of osmoregulation does not appear to be proportional to the increase in osmotic gradient in 30‰ salinity.

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Introduction

While the mechanisms involved in the ionic and osmotic regulation of internal body fluids have been extensively studied, e.g. in the genera *Salmo* and *Oncorhynchus* (Huntsman and Hoar 1939; Black 1951; Parry 1958, 1960; Gordon 1959, 1963; Houston 1959, 1960, 1961; Houston and Threadgold 1963), relatively few investigations have been made of the relationship between metabolism and osmoregulation in fishes except the preliminary study of Hickman (1959) on the flounder, *Platichthys stellatus*. The present study has been undertaken to provide information on the cost of osmoregulation of rainbow trout, *Salmo gairdneri*, at various levels of activity after acclimation to different salinities and temperatures.

The rainbow trout is a euryhaline teleost which survives well in both fresh water and the sea. It has been described as a good osmoregulator by Busnel (1942), Houston (1959 and 1961), Parry (1960 and 1961), and Gordon (1963). Conte and Wagner (1965) have investigated the seasonal changes in ionic and osmoregulatory ability of the steelhead trout. They came to the conclusion that the fish were near the peak of seawater adaptation at the onset of the migratory period (April–May) and return to a condition of

being nearly stenohaline by early June. The measurements on fish acclimated to seawater in this study were carried out in March, April, and May, 1965 and 1966.

Materials and Methods

Rainbow trout ranging in weight from 23 to 196 g were obtained from the stock kept at the Laboratory for Experimental Limnology, Maple, Ontario. The fish were held in tanks of 100-gal capacity with continuous recirculation of water through gravel filters. They were fed a commercial food in pellet form daily with a weekly supplement of beef liver. No prophylactic measures were taken. The fish were acclimated to fresh water and to dilutions of artificial seawater to provide salinities of 7.5‰, 15‰, and 30‰. Seawater was prepared by the addition of synthetic salt (supplied by Aquarium Systems Inc., Ohio) to dechlorinated Toronto tap water. In acclimating the fish to the desired temperature (5° and 15 °C) a daily change of 1 °C was followed. The fish were moved to the desired salinity in steps of 7.5‰ at approximately 2-week intervals. After the prescribed temperature and salinity had been reached the fish were allowed to remain for at least 2 weeks before the experiments were started. The fish were tested at the respective acclimation temperatures and salinities. The salinities were periodically checked by the method of Harvey (1928) and maintained $\pm 1\%$. The temperature was maintained at ± 0.5 °C. A 12-hour light period was maintained throughout.

Standard oxygen consumption was determined with Fry's apparatus (Smit 1965) in which simultaneous measurements of spontaneous activity and oxygen consumption were obtained. A linear relation between these values exists when plotted on a semi-logarithmic grid. The standard oxygen consumption was estimated by extrapolation to zero activity of the straight line fitted according to the method of least squares as in Fig. 1. A fish was in the respirometer for at least 12 hours before the experiment was

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started and was not fed for at least 24 hours before that; thus the elevation of standard metabolism caused by feeding was eliminated (Beamish 1964). After measurement of standard oxygen consumption, the fish was transferred to a recirculating tubular respirometer originally designed by Blazka (see description in Smit 1965) for studies on active oxygen consumption. Again the fish was left in the apparatus for at least 12 hours before measurements were begun. Active oxygen consumption was determined at velocities of 18.5, 45.1, 57.5, or 72.7 cm/sec. Finally current speed was increased in steps up to the maximum value at which the fish could maintain steady swim-

ming indefinitely. As Kutty (1966) showed, the rainbow trout does not require any training to swim steadily against the current.

The first measurement of oxygen consumption at a given current speed was made during the second half hour of swimming. For the first half hour aerated water flowed through the chamber. A water sample was then taken at the end of the first half hour, and the flow of water through the chamber was immediately cut off. The second water sample, required for determination of the amount of oxygen consumption, was withdrawn from the chamber at the end of the second half hour. The chamber was then opened

TABLE I

Influence of temperature and salinity on oxygen consumption. Equations for relation between metabolic rate ($\bar{Y}$, mg O_2 /h) and weight (X , g) at different levels of activity in various salinities

Salinity, ‰	Activity, cm/sec	No. fish	Weight range, g	Regression equation, $\log Y = a + b \log X$	S.D. of slope, $\pm$
Temperature, 5 °C					
Fresh water	0	17	33-150	$\log Y = -0.8108 + 0.7838 \log X$	0.073
	18.5	17	33-150	$\log Y = -0.5460 + 0.7836 \log X$	0.016
	45.1	17	33-150	$\log Y = -0.3362 + 0.7909 \log X$	0.044
	57.5	17	33-150	$\log Y = -0.3551 + 0.8564 \log X$	0.039
	Maximum	17	33-150	$\log Y = -0.5257 + 1.0437 \log X$	0.043
7.5‰	0	17	27-153	$\log Y = -0.7310 + 0.7345 \log X$	0.035
	18.5	17	27-153	$\log Y = -0.4913 + 0.7197 \log X$	0.027
	45.1	17	27-153	$\log Y = -0.3942 + 0.7614 \log X$	0.035
	57.5	17	27-153	$\log Y = -0.4371 + 0.8421 \log X$	0.040
	Maximum	17	27-153	$\log Y = -0.6694 + 1.0465 \log X$	0.048
15‰	0	17	28-149	$\log Y = -0.9541 + 0.8682 \log X$	0.063
	18.5	17	28-149	$\log Y = -0.4839 + 0.7557 \log X$	0.038
	45.1	17	28-149	$\log Y = -0.3811 + 0.8184 \log X$	0.045
	57.5	17	28-149	$\log Y = -0.4094 + 0.9017 \log X$	0.062
	Maximum	17	28-149	$\log Y = -0.5520 + 1.0552 \log X$	0.062
30‰	0	14	25-196	$\log Y = -0.3738 + 0.6234 \log X$	0.032
	18.5	14	25-196	$\log Y = -0.3461 + 0.7075 \log X$	0.042
	45.1	14	25-196	$\log Y = -0.3297 + 0.8099 \log X$	0.046
	57.5	14	25-196	$\log Y = -0.4475 + 0.9389 \log X$	0.034
	Maximum	14	25-196	$\log Y = -0.3183 + 0.9590 \log X$	0.060
Temperature, 15 °C					
Fresh water (0.2‰)	0	16	54-135	$\log Y = -0.5147 + 0.7834 \log X$	0.072
	18.5	16	54-135	$\log Y = -0.3362 + 0.7829 \log X$	0.104
	45.1	16	54-135	$\log Y = -0.3496 + 0.8982 \log X$	0.112
	72.7	16	54-135	$\log Y = -0.2235 + 0.9337 \log X$	0.096
	Maximum	16	54-135	$\log Y = -0.2146 + 0.9916 \log X$	0.068
7.5‰	0	15	47-138	$\log Y = -0.7088 + 0.8408 \log X$	0.122
	18.5	15	47-138	$\log Y = -0.3442 + 0.7332 \log X$	0.078
	45.1	15	47-138	$\log Y = -0.5642 + 0.9601 \log X$	0.057
	72.7	15	47-138	$\log Y = -0.1096 + 0.8280 \log X$	0.109
	Maximum	15	47-138	$\log Y = -0.6290 + 1.1581 \log X$	0.072
15‰	0	14	24-198	$\log Y = -0.5115 + 0.7743 \log X$	0.054
	18.5	14	24-198	$\log Y = -0.3480 + 0.7940 \log X$	0.070
	45.1	14	24-198	$\log Y = -0.2148 + 0.8218 \log X$	0.071
	72.7	14	24-198	$\log Y = -0.1048 + 0.8758 \log X$	0.067
	Maximum	14	24-198	$\log Y = -0.3250 + 1.0500 \log X$	0.072
30‰	0	19	23-138	$\log Y = -0.1407 + 0.6226 \log X$	0.066
	18.5	19	23-138	$\log Y = -0.0311 + 0.6249 \log X$	0.028
	45.1	19	23-138	$\log Y = -0.2134 + 0.8558 \log X$	0.047
	72.7	19	23-138	$\log Y = -0.3337 + 1.0059 \log X$	0.034
	Maximum	19	23-138	$\log Y = -0.5883 + 1.1919 \log X$	0.042

again to the flow of air-saturated water for the next half hour. This protocol of half-hourly alternation between flushing the chamber with a current of freshly aerated water and recirculating the same water was continued until a constant rate of oxygen consumption was measured for two consecutive test periods. Each fish swam steadily for at least 10 hours, the speed being increased stepwise to each rate indicated above.

At the end of the experiment the fish was weighed and its fork length measured to the nearest millimeter.

Oxygen dissolved in the water sample was determined with a dropping mercury electrode (Southern Analytical W.P.R.L. A1672 Dissolved Oxygen meter). This instrument was checked daily by the unmodified Winkler's method (American Public Health Association 1955) and recalibrated if necessary.

Results and Discussion

For a given activity, oxygen consumption increases linearly with body weight when plotted on a double logarithmic grid (Fig. 2). Equations for the regression of log metabolic rate on log body weight were calculated for each series of data and are given in Table I. For the present purposes the metabolic rate of a 100-gram fish was calculated (Table II). In Fig. 3 the logarithms of the metabolic rates of a 100-g fish are plotted against the

swimming speed. The standard rate shown was obtained with Fry's respirometer while active rates were obtained with Blazka's respirometer. As can be seen in Fig. 3 the values of extrapolation to zero speed, as in Brett (1965), and the standard values obtained separately in Fry's respirometer are similar. The fish exhibit the lowest rates of oxygen consumption at all levels of activity in 7.5‰ salinity. In fresh water and in 15‰ salinity the rates are approximately equal to each other. The oxygen consumption in 30‰ salinity is highest at all levels of activity.

Two sets of differences have been calculated between the values for respiration in Table II. The series interdigitated vertically show the difference between the metabolic rate at a given level of activity and standard metabolism in the respective salinity. The horizontal series show the differences in metabolism at a given level of activity between 7.5‰, a salinity which is approximately isosmotic with the blood of rainbow trout (Rao, in press) and the three other salinities (fresh water, 15‰, and 30‰). It is presumed that the vertical series of differences represent the combined cost of swimming and osmoregulation and that the horizontal

TABLE II

Oxygen consumption in mg/kg per hour of a 100-g fish acclimated to various salinities and temperatures, and tested at different levels of activity

SPEED cm/sec	F.W.	7.5‰	15‰	30‰
0	57	54	60	74
18.5	105	89	106	117
45.1	176	134	180	195
57.5	228	176	247	269
MAXIMUM	354	266	360	398
15°				
0	112	94	109	127
18.5	169	132	172	191
45.1	279	227	269	315
72.7	436	352	443	476
MAXIMUM	584	486	597	624

NOTE. The metabolic rates are calculated from the regression equations given in Table I. The small numbers are differences as explained below. The series interdigitated vertically show the difference between the metabolic rate at a given level of activity and standard metabolism (for each salinity). The horizontal series show the difference in metabolism at a given level of activity between the lowest at 7.5‰ and the three other salinities (fresh water, 15‰, and 30‰).

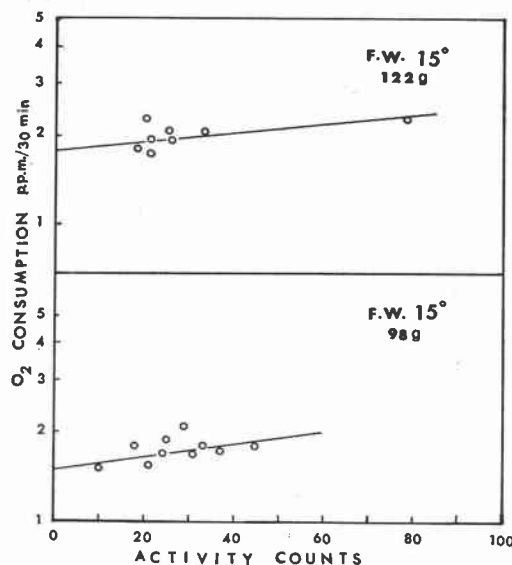


FIG. 1. Two examples of the relation between activity and oxygen consumption using Fry's respirometer. The rate of oxygen consumption at zero activity determined by extrapolation is the standard rate.

series represent the cost of osmoregulation at the respective levels of activity and osmotic stress.

It is considered that a test of the assumption given above is to subtract the cost of osmoregulation from the combined cost of swimming and osmoregulation to see if the cost of swimming is then independent of the salinity of the medium. As is indicated in

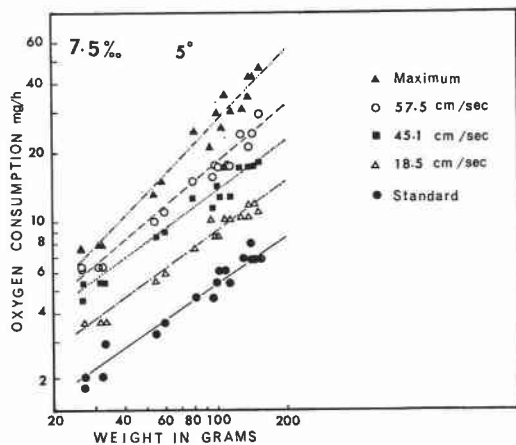


FIG. 2. Oxygen consumption as a function of weight for fish acclimated to 5°C in 7.5‰ salinity at various levels of activity.

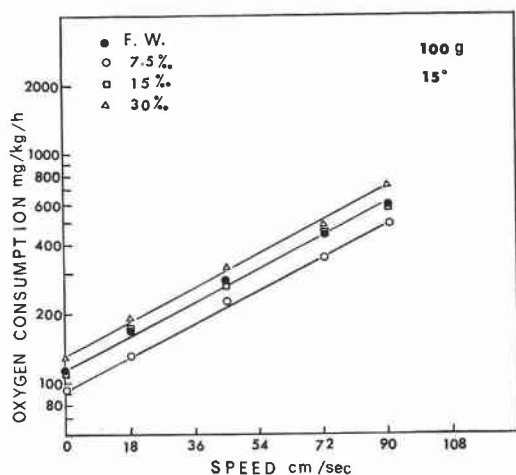


FIG. 3. The relation between swimming speed and the rate of oxygen consumption of a 100-g rainbow trout acclimated to and tested in freshwater, 7.5‰ and 30‰ salinity at 15°C. The standard rate of oxygen consumption determined by extrapolation to zero activity of the straight line fitted to the data agrees with the value obtained separately with Fry's respirometer.

Fig. 4, where such differences are plotted, the cost of swimming at a given temperature does appear to be independent of salinity. The scatter is large at a current speed of 18.5 cm/sec, but a large degree of error is associated with these points. It may therefore be concluded that the differences between the metabolic rates at a given level of activity in various salinities can be ascribed to the cost of osmoregulation. Such differences are plotted against the respective total cost of metabolism in Fig. 5.

It can be seen from Fig. 5 that the cost of osmoregulation in a given medium is nearly proportional to the metabolic rate, which, in turn, is correlated with the rate of irrigation of the gills (Saunders 1962). Approximately 20% of total metabolism is involved in the process of osmoregulation in fresh water, whereas it is about 27% in 30‰ salinity. The cost of osmoregulation on either side of the isosmotic point (7.5‰) is the same for the same gradient (in fresh water and 15‰ salinity). The increase in the cost of osmoregulation in 30‰ salinity is not proportional to the increase in osmotic gradient of the medium when compared to fresh water. Osmotic concentration does not remain constant with activity (Rao, ms). In the rainbow trout the internal concentration is somewhat higher in exercised fish than in resting ones. However, the difference is not sufficient to account for the deviation from linearity of the relation of the cost of osmoregulation to the osmotic gradient. It is also interesting to note that the cost of osmoregulation at 5°C appears to be somewhat higher than that at 15°C both in fresh water and 30‰ salinity.

It was suggested by Gordon (1963) that the most important process of salinity acclimation in rainbow trout is a reduction in integumentary permeability to water. He concluded that the adaptation to different salinities does not necessarily impose on rainbow trout markedly different energy requirements for osmoregulatory purposes and that much of the adjustment is taken care of by changes in the physical state of the permeable areas of the integument changes perhaps mediated by "a neurohypophysial

hormone". This might explain the reason for lack of proportionality in the cost of osmoregulation in relation to osmotic gradient.

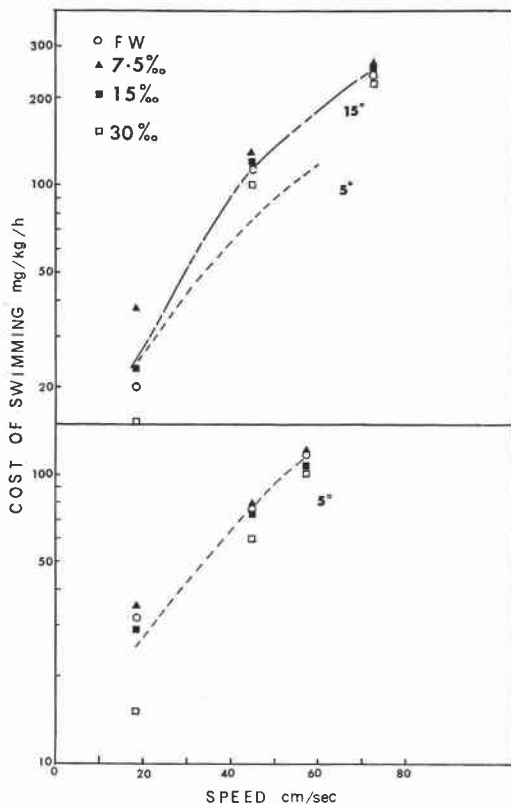


FIG. 4. Cost of swimming in a 100-g fish at a given temperature (5° and 15°C). For explanation see text.

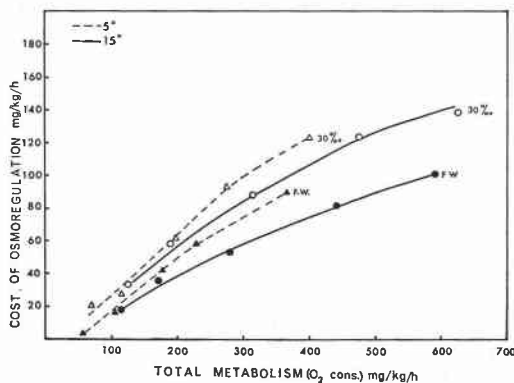


FIG. 5. Relation between total metabolism and the cost of osmoregulation in fresh water and 30‰ salinity for a 100-g fish. The cost for 15‰ salinity is the same as in fresh water. For explanation see text.

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